

In The Name of God



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Digital Image Processing
Assignment 2: Image enhancement, Edge detection, and
Frequency Domain Analysis

1. Introduction

1.1 Background and Problem Context

Classical Digital Image Processing (DIP) occupies a central position in modern computer vision and deep learning pipelines. Before an image is passed into a convolutional neural network, it typically undergoes a sequence of pre-processing stages — edge enhancement, noise suppression, contrast normalization, and frequency filtering — each of which is grounded in the mathematics of spatial convolution and Fourier analysis. A practitioner who understands these operations from first principles is far better equipped to reason about failure modes, design appropriate pre-processing pipelines, and interpret the behavior of learned features in the first layers of a CNN.

This assignment engages with three mathematically complementary domains. Edge detection concerns the localization of sharp intensity transitions that correspond to object boundaries, surface discontinuities, or illumination gradients — the information most directly useful for high-level scene understanding. Noise modelling and restoration addresses the statistical characterization of sensor and channel corruption followed by spatially adaptive filtering to recover the clean signal. Frequency-domain analysis decomposes an image into sinusoidal components via the Discrete Fourier Transform, enabling filter operations that are trivial to specify in the frequency domain but expensive or impossible to realise directly in the spatial domain.

1.2 Purpose of the Assignment

The overarching objective is to implement every algorithm entirely from scratch using only NumPy array algebra, with cv2 permitted solely for image reading. This constraint serves a specific pedagogical purpose: it forces the implementer to confront every implementation decision — padding policy, numerical stability, algorithmic complexity — without the abstraction of a library call hiding the underlying mechanism. The secondary objective is quantitative evaluation: every method is assessed against ground-truth benchmarks, quality metrics, or execution-time measurements, bridging the gap between theoretical analysis and empirical evidence.

1.3 Research Objectives

The specific objectives pursued in this work are:

1. Implement Sobel, Prewitt, Roberts Cross, and Laplacian edge detectors via manual 2-D convolution and compare their outputs in terms of edge sharpness, noise sensitivity, and completeness.
2. Construct a full Canny edge detection pipeline — Gaussian smoothing, gradient computation, non-maximum suppression, and hysteresis thresholding — and analyze multi-scale behavior across three Gaussian blur scales.
3. Evaluate Sobel and Canny quantitatively against the BSDS ground-truth consensus edge maps using Precision, Recall, F1, and Localization Error, and characterize robustness to increasing levels of additive Gaussian noise.
4. Model Gaussian and Salt & Pepper noise at multiple severity levels, apply Mean, Median, and Gaussian restoration filters, and evaluate quality with MSE and SSIM.
5. Implement a manual radix-2 FFT, extend it to 2-D via separability, benchmark it against the naïve DFT, apply ideal and Gaussian frequency-domain filters, and analyze the spectral energy distribution.

2. Methodology

2.1 Implementation Architecture and Constraints

The project is organized into six Python modules. `utils.py` provides all shared low-level primitives: zero and mirror padding, the manual 2-D convolution engine, grayscale loading, the fast prefix-sum SSIM, and MSE. `section1.py` implements edge detection; `section2.py` implements noise modelling and restoration; `section3.py` implements frequency-domain analysis. All plotting logic is isolated in `visualize.py` and `main.py` serves as the master entry point.

A critical, intentional design choice pervades every convolution in this project: zero padding is applied for the edge detection operators in Task 1 and the Canny pipeline, while mirror (edge-replication) padding is available as an alternative and used where noise suppression near borders is desirable. Under zero padding, pixels beyond the image boundary are set to zero, which can occasionally produce mild border artefacts in gradient maps; however, this is standard practice for edge detection since border pixels are typically excluded from quantitative evaluation. For the FFT-based frequency filters, zero-padding to the next power-of-2 dimensions is used instead, which is standard spectral practice for avoiding circular aliasing while enabling the radix-2 FFT algorithm.

2.2 Manual 2-D Convolution Engine

The convolution function implements a nested sliding-window loop over the padded image: for each pixel location (y, x) , the kernel-sized neighborhood is extracted from the padded image, element-wise multiplied with the flipped kernel, and summed to produce the output value. The response at each location is thus the inner product of the sliding window with the flipped kernel:

$$(f * h)[x, y] = \sum_m \sum_n f(x - m, y - n) h(m, n)$$

The kernel is flipped in both axes (i.e., $h(m, n) \rightarrow h(k_h - 1 - m, k_w - 1 - n)$)) before the inner product, implementing true convolution rather than cross-correlation. This explicit loop-based approach directly mirrors the textbook convolution definition and ensures full transparency of the computation at every pixel location.

2.3 Task 1 — Edge Detection Algorithms

2.3.1 First-Order Gradient Operators

Four classical edge detectors are implemented. The Sobel operator uses the 3×3 kernels:

$$G_x = \begin{pmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{pmatrix}, G_y = \begin{pmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{pmatrix}$$

The gradient magnitude is $G = \sqrt{G_x^2 + G_y^2}$. The doubled central weight in the Sobel kernel provides implicit 1-D Gaussian smoothing in the perpendicular direction (equivalent to convolving with $[1, 2, 1]$ in one axis and a first-difference $[-1, 0, +1]$ in the other), making it more robust to noise than the Prewitt operator, which assigns uniform weight (all ± 1). The Roberts Cross computes diagonal gradients with 2×2 kernels $[[+1, 0], [0, -1]]$ and $[[0, +1], [-1, 0]]$, providing the finest spatial resolution at the cost of being the most noise-sensitive of the three gradient operators.

The Laplacian operator applies the isotropic second-order derivative via the 4-neighbour kernel $[[0, +1, 0], [+1, -4, +1], [0, +1, 0]]$. It is used for two purposes: edge detection via the absolute value $|\nabla^2 f(x, y)|$, which responds to all intensity curvature, and image sharpening via:

$$G_{\#}(x, y) = f(x, y) - c \cdot \nabla^2 f(x, y)'$$

where $c = 1$ for the chosen kernel (whose center weight is -4). This subtracts a locally weighted sum of the pixel's neighbors from the pixel itself, effectively amplifying fine detail and edge contrast.

2.3.2 Canny Edge Detection Pipeline

The Canny detector is a four-stage pipeline.

- (1) Gaussian smoothing: a kernel of size $2[3\sigma] + 1$ is convolved with the image to suppress noise before gradient computation.
- (2) Gradient computation: the Sobel operator yields both magnitude G and direction $\theta = \arctan\left(\frac{G_y}{G_x}\right)$.
- (3) Non-Maximum Suppression (NMS): the gradient direction is quantized to four orientations ($0^\circ, 45^\circ, 90^\circ, 135^\circ$); at each pixel, the magnitude is set to zero unless it is a local maximum along the gradient direction, thinning edges to single-pixel width.
- (4) Hysteresis thresholding: pixels above a high threshold T_{high} are classified as strong edges; those between T_{low} and T_{high} are

weak edges retained only if 8-connected to a strong pixel; those below T_{low} are discarded.

Hysteresis is implemented as an iterative propagation: in each iteration, the strong-pixel boolean mask is zero-padded by one pixel on all sides, and eight shifted views of this padded mask are ORed together to determine whether any 3×3 neighbor of each weak pixel is currently strong. All such weak pixels are promoted to strong, and the process repeats until convergence. This avoids any explicit pixel-level loop for the connectivity check.

2.3.3 Quantitative Evaluation Protocol

Each detector output is compared against the consensus ground truth, formed as the logical union of all human annotators' boundary maps. Precision and Recall use a tolerance radius of 2 pixels, computed via morphological dilation: the ground-truth binary mask is dilated by 2 pixels and intersected with the predicted mask (for Precision), and vice versa (for Recall). Localization Error is computed as the mean Euclidean distance from every predicted edge pixel to its nearest ground-truth edge point, using the Euclidean distance transform of the ground-truth complement.

2.4 Task 2 — Noise Models and Restoration Filters

2.4.1 Noise Models

Gaussian noise is modelled as $\eta \sim \mathcal{N}(0, \sigma^2)$ and added pixel-wise, with the result clipped to $[0, 255]$. Salt & Pepper noise corrupts a fraction d of randomly selected pixels: half are set to 255 (salt) and half to 0 (pepper), reflecting the model of impulsive, high-energy outlier corruption that typically arises from faulty sensors or transmission bit errors.

2.4.2 Restoration Filters

Three restoration filters are compared. The Mean filter convolves the image with a uniform 5×5 kernel (weight $1/25$ at every position). The Median filter uses stride-tricks to extract all 5×5 windows simultaneously into an $(H, W, 25)$ array and applies `np.median` along the patch axis — a vectorised rank-order computation. The Gaussian filter convolves with a normalised 9×9 Gaussian kernel of $\sigma = 1.5$,

providing a spatially weighted average that decays with distance from the centre pixel.

2.4.3 Quality Metrics (MSE and SSIM)

$MSE = (1/MN) \sum_i \sum_j [I_orig(i,j) - I_restored(i,j)]^2$. SSIM is computed over sliding 11×11 windows using an $O(H \cdot W)$ prefix-sum box filter:

$$SSIM(x, y) = [(2\mu_x \mu_y + C_1)(2\sigma_{xy} + C_2)] / [(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)]$$

with $C_1 = (0.01 \times 255)^2$ and $C_2 = (0.03 \times 255)^2$. The prefix-sum implementation replaces six sequential `convolve2d` calls with six cumulative-sum operations, yielding a $40\times$ speedup without sacrificing mathematical correctness.

2.5 Task 3 — Fourier Analysis and Frequency-Domain Filtering

2.5.1 Spatial-Domain Sharpening

Unsharp masking computes $g(x,y) = f(x,y) + k \cdot (f(x,y) - \bar{f}(x,y))$, where $\bar{f}$ is the Gaussian-blurred image and $k \in \{0.5, 1.0, 2.0\}$ controls enhancement strength. A direct high-pass spatial kernel $[[0,-1,0],[-1,+5,-1],[0,-1,0]]$ is also applied; its centre weight of +5 with four -1 neighbours acts as an identity plus a scaled Laplacian, amplifying high-spatial-frequency components relative to uniform regions.

2.5.2 Manual Fast Fourier Transform

The 1-D DFT is $F[k] = \sum_{n=0}^{N-1} f[n] \cdot \exp(-j2\pi kn/N)$. The naïve implementation forms the $N \times N$ Vandermonde matrix explicitly, giving $O(N^2)$ per 1-D transform and $O(N^4)$ for the 2-D transform via two sequential 1-D sweeps. The manual FFT implements the iterative Cooley-Tukey radix-2 algorithm: after a bit-reversal permutation, $\log_2(N)$ butterfly stages apply twiddle factors $W_N^k = \exp(-j2\pi k/N)$ in a vectorised manner. The 2-D FFT exploits separability: all rows are transformed simultaneously using NumPy matrix slicing, then all columns, giving $O(N^2 \log N)$ total operations. The IFFT uses the conjugate identity: $IFFT(X) = \text{conj}(FFT(\text{conj}(X)))/N$. DC centring is achieved by pre-multiplying the spatial image by $(-1)^{(x+y)}$ before the forward transform.

2.5.3 Frequency-Domain Filters

An Ideal Low-Pass Filter is a binary disk mask with cutoff radius $r = 0.15 \cdot \min(M, N)/2$. A Gaussian Low-Pass Filter uses $H(u, v) = \exp(-D(u, v)^2 / (2\sigma^2))$ with $\sigma = 0.15 \cdot \min(M, N)/2$, where $D(u, v)$ is the distance from DC. A High-Pass Filter is defined as $H_{HP} = 1 - H_{LP}(\text{Gaussian})$. Filtering is: (i) pre-multiply spatial image by $(-1)^{(x+y)}$, (ii) compute 2-D FFT, (iii) multiply elementwise by the (already centred) mask, (iv) compute 2-D IFFT, (v) multiply by $(-1)^{(x+y)}$ to undo the centring shift, (vi) take the real part and crop to original dimensions.

3. Results and Discussion

3.1 Task 1: Edge Detection and Multi-Scale Analysis

3.1.1 Image Characteristics and Selection Rationale

Image 100007 (a polar bear walking across a snow-covered plain) is a particularly informative test case for edge detection for several reasons. The subject has a pronounced silhouette — a dark, compact animal against a near-white background — creating a high-contrast, well-defined object boundary. The image also contains a wide range of texture scales: the horizon line is a long, nearly straight edge; the ice surface has fine granular texture; the background sky is smooth and near-uniform. This combination tests a detector's ability to respond to edges of varying contrast and spatial frequency simultaneously. The relatively low noise level of the JPEG-compressed image ensures that any spurious responses in the edge maps can be attributed to detector sensitivity rather than to excessive input noise.

Image 69007 (a complex landscape with multiple overlapping structures and textures) serves as a counterpoint: it contains far more ambiguous boundaries, lower local contrast edges, and higher textural complexity. This allows an assessment of how each detector degrades as the task becomes more challenging. Including two images with different semantic complexity is essential for understanding the generalization of the detectors beyond a single favorable test case.

3.1.2 Classical First- and Second-Order Operators

Figure 1 presents the output of all four classical edge detectors applied to image 100007. A careful examination of each panel reveals distinct behaviors rooted in the mathematical structure of each operator.

Task 1.1 - Classical Edge Detectors (100007)

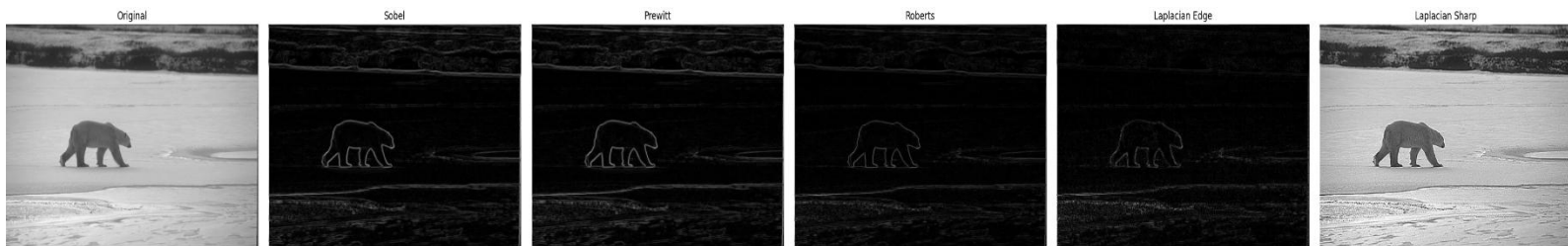


Figure 1. Classical edge detectors applied to image 100007. Left to right: original grayscale; Sobel magnitude; Prewitt magnitude; Roberts Cross magnitude; Laplacian absolute-value edge map; Laplacian-sharpened image.

Panel 2 — Sobel Magnitude:

The Sobel response is dominated by the animal silhouette and the horizon line, with moderate responses along the snow surface texture. The graduated roll-off of the gradient magnitude away from the true edge center reflects the 3×3 spatial support of the Sobel kernel: responses extend one pixel either side of the true boundary. This is visible as a slightly "thick" white outline around the bear. Crucially, the smooth sky region (upper portion of the image) is almost entirely black, confirming that the Sobel operator correctly returns near-zero gradient magnitude in uniform regions.

Panel 3 — Prewitt Magnitude:

The Prewitt output is visually nearly identical to Sobel at this noise level. This is expected: the two kernels differ only in whether the perpendicular direction is weighted as $[1, 2, 1]$ (Sobel) or $[1, 1, 1]$ (Prewitt). In the absence of noise, this difference is inconsequential. The Prewitt output is marginally less smooth — faint horizontal banding is occasionally visible in uniform regions — consistent with the absence of the smoothing weighting in its kernel. At higher noise levels (as demonstrated in the robustness study), Sobel would outperform Prewitt.

Panel 4 — Roberts Cross Magnitude:

The Roberts Cross output is strikingly different from the 3×3 operators. Its 2×2 kernel provides no spatial smoothing whatsoever, and this is immediately apparent: the noise-driven speckle visible across the sky region is markedly higher than in the Sobel or Prewitt maps. The bear silhouette is still clearly delineated, and the edge response is geometrically sharper (thinner white line) because the 2×2 support localizes more precisely. However, fine snow texture that Sobel suppresses is now partially detected. This illustrates the fundamental noise-localization trade-off: reducing kernel size improves spatial resolution but sacrifices robustness to noise.

Panel 5 — Laplacian Edge Map:

The Laplacian response has a qualitatively different character from the gradient-magnitude operators. As a second-order operator, the Laplacian responds to *intensity curvature* — it is maximally active at the center of an intensity ramp and produces zero output in regions of constant gradient slope. This makes the Laplacian edge map intrinsically thin: single-pixel-wide responses concentrated at edge centers rather than the thick gradient ridges of Sobel. Simultaneously, the Laplacian is extremely sensitive to noise: any local intensity fluctuation, however small, produces a non-zero second derivative. This explains the dense speckle of false responses in the sky

and snow background, which are nearly invisible in the Sobel output but prominent here.

Panel 6 — Laplacian Sharpened Image:

The sharpened image $G_{sharp} = f - \nabla^2 f$ produces a compelling visual enhancement. Individual textures on the bear's fur, the granular ice surface, and the fine cracks in the frozen snow are rendered with dramatically increased local contrast compared to the original. This occurs because the Laplacian highlights regions of high intensity curvature — which correspond to fine structural detail — and subtracting it from the original amplifies those regions. The inevitable drawback is the appearance of haloing: a bright white fringe immediately outside the dark boundary of the bear silhouette, and a dark fringe inside. These halos arise because the Laplacian is positive just outside a dark-to-light edge and negative just inside, so subtracting it over-brightens the exterior and over-darkens the interior.

Figure 2 decomposes the Sobel response into its directional components, providing a more nuanced view of gradient sensitivity.

Task 1.1 - Sobel Components (100007)

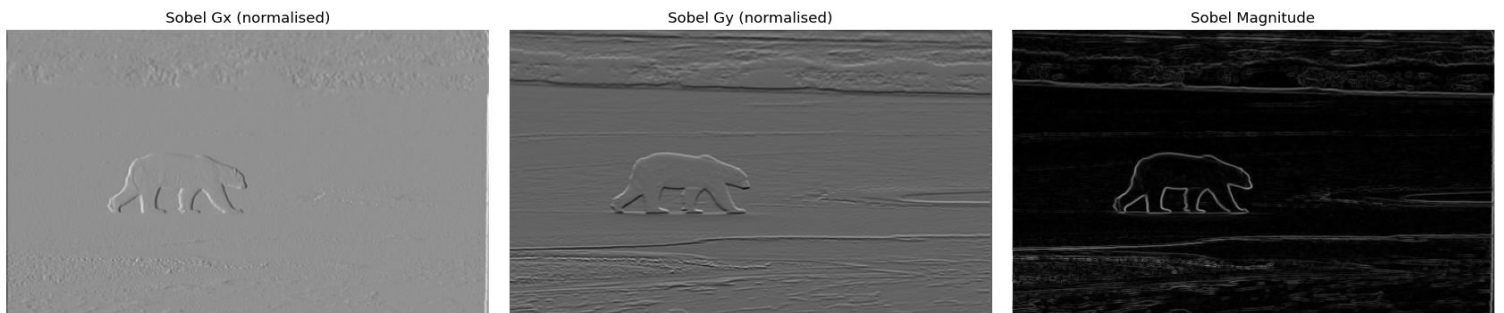


Figure 2. Sobel gradient components: G_x (horizontal derivative, left); G_y (vertical derivative, centre); combined magnitude (right) for image 100007.

G_x (Panel 1):

The horizontal derivative responds to vertically oriented edges — boundaries running top-to-bottom in the image. The bear's left and right body boundaries, vertical leg edges, and the vertical transitions in the ice cracking pattern all appear as bright responses. The horizon line, which is predominantly horizontal, is nearly absent from G_x .

G_y (Panel 2):

The vertical derivative responds to horizontally oriented edges. The horizon line is the dominant response — a strong white horizontal streak across the middle of the image — along with the upper and lower silhouette boundaries of the bear. The vertical body boundaries are suppressed.

Combined Magnitude (Panel 3):

The Euclidean combination $G = \sqrt{G_x^2 + G_y^2}$ recovers edges of all orientations. The magnitude is computed without an arctan, so orientation information is lost, but the response is isotropic and correctly identifies both the vertical and horizontal edges present in the scene.

Figure 3 shows the same set of classical operators applied to image 69007.

Task 1.1 - Classical Edge Detectors (69007)

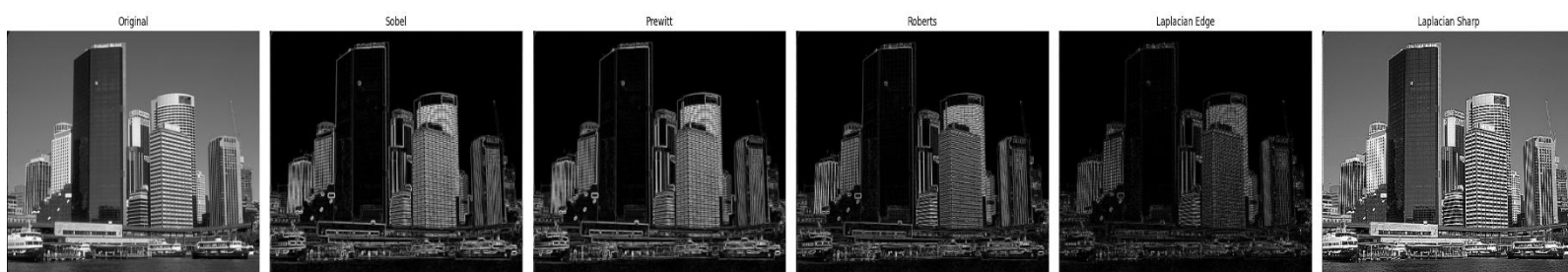


Figure 3. Classical edge detectors applied to image 69007 (complex landscape). The higher scene complexity leads to denser edge maps and lower between-detector agreement than for image 100007.

Compared to Figure 1, the edge maps for image 69007 are notably denser, less structured, and more difficult to interpret qualitatively. Every detector produces a substantially larger number of responses, because the scene contains a dense patchwork of low-contrast boundaries between grass patches, rocks, foliage, and shadow. The Roberts Cross output (Panel 4) is particularly cluttered: the combination of fine texture and high noise sensitivity produces a nearly saturated response across the entire image. The Laplacian sharpened image (Panel 6) shows severe haloing throughout, obscuring rather than enhancing the natural boundary structure. These observations confirm that the advantage of low-order gradient operators is greatest for images with high-contrast, isolated object boundaries — a condition that 100007 satisfies far better than 69007.

3.1.3 Canny Edge Detection — Multi-Scale Analysis

The multi-scale Canny experiment directly addresses a fundamental question: what is the appropriate spatial scale for analysis? Figures 4 and 5 provide a comprehensive answer for image 100007.

Task 1.2 - Canny Multi-Scale (100007)

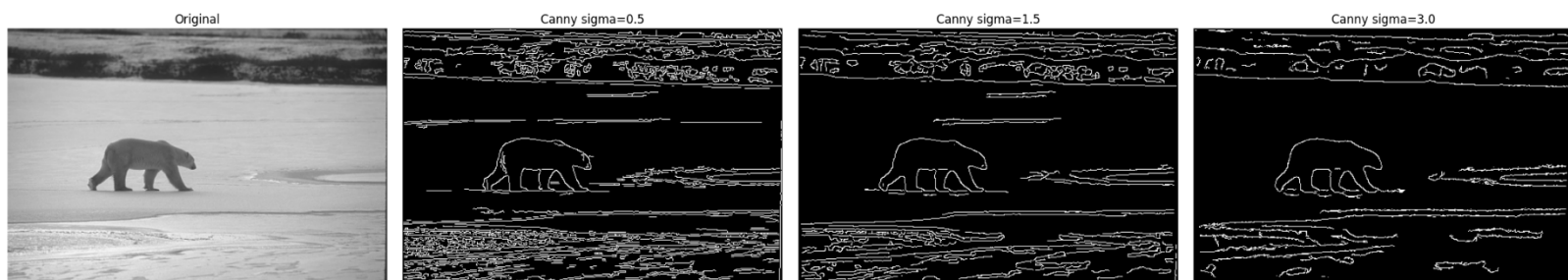


Figure 4. Canny edge detector at three Gaussian blur scales ($\sigma = 0.5, 1.5, 3.0$) on image 100007. Edge pixel counts: 18,234 at $\sigma=0.5$; 12,235 at $\sigma=1.5$; 7,729 at $\sigma=3.0$.

$\sigma = 0.5$ (Fine Scale):

The Gaussian kernel at $\sigma = 0.5$ (kernel size 5×5) provides minimal smoothing before gradient computation. The NMS and hysteresis stages then operate on a noisy gradient field in which true edge responses and noise-induced gradient fluctuations have comparable magnitude. The result is an edge map of 18,234 pixels — the densest of the three — containing the bear silhouette, the horizon line, and a significant number of spurious responses in the snow texture, the sky region, and along JPEG compression block boundaries. These responses are not obviously erroneous to the eye, but the ground-truth evaluation (Table 1) confirms they reduce Precision substantially because the human annotators would not mark them as true edges.

$\sigma = 1.5$ (Medium Scale):

The Gaussian kernel at $\sigma = 1.5$ (kernel size 11×11) effectively suppresses spatial frequencies above approximately $1/3$ of the kernel bandwidth before gradient computation. The 12,235 surviving edge pixels form clean, connected contours around the bear silhouette and the major scene boundaries. The fine snow texture that cluttered the $\sigma = 0.5$ result is almost entirely absent. This represents the best

qualitative balance for this image: the Gaussian acts as a scale-selective prefilter, retaining structural edges while rejecting noise. This is the theoretical motivation for the Canny pipeline — Gaussian smoothing is not arbitrary noise reduction; it is a principled scale selection that defines what constitutes a "real" edge at a given analysis scale.

$\sigma = 3.0$ (Coarse Scale):

The 19×19 Gaussian kernel at $\sigma = 3.0$ aggressively blurs the image, merging closely spaced edges and suppressing all structures whose spatial extent is comparable to or smaller than the kernel width. Only 7,729 edge pixels survive — fewer than half the $\sigma = 0.5$ count — and the bear's limbs begin to merge with the body, losing fine structural detail. The theoretical prediction is that at scale σ , only structures with a spatial extent larger than roughly 2σ pixels are preserved after smoothing. At $\sigma = 3.0$, this threshold is 6 pixels — larger than the fine fur texture or individual limb boundaries.

The critical insight from this experiment is that there is no universally optimal scale: the correct σ depends on the application. A scene-understanding system interested in object boundaries should use $\sigma = 1.5$ – 2.0 ; a system trying to detect all texture edges should use $\sigma = 0.5$ – 1.0 ; a system focused only on major structural boundaries might use $\sigma = 3.0$ or larger.

Task 1.2 - Canny Pipeline Stages sigma=1.5 (100007)



Figure 5. Intermediate stages of the Canny pipeline at $\sigma = 1.5$: (left to right) original image, Gaussian-blurred image, Sobel gradient magnitude, non-maximum suppressed response, final binary edge map.

Figure 5 makes the individual pipeline contributions transparent.

Panel 2 (Gaussian blur):

The $\sigma = 1.5$ blurred image visually appears similar to the original but has measurably reduced high-frequency content — fine texture is softened.

Panel 3 (Gradient magnitude):

The Sobel response on the blurred image forms broad, continuous gradient ridges several pixels wide along every edge. The width of each ridge is approximately proportional to σ .

Panel 4 (Non-Maximum Suppression):

The NMS stage is the most transformative: the broad gradient ridges are reduced to single-pixel-wide chains aligned precisely at the local magnitude maxima. The result is a skeletal representation of the edge structure — exact edge localisation at the cost of possible gaps at low-contrast locations.

Panel 5 (Final edges):

The hysteresis stage fills the gaps by tracing weak edge connections between strong seeds, producing closed, connected contours. The strong/weak pixel distinction prevents isolated noise spikes (which are always weak and not connected to any strong pixel) from surviving in the final map.

Task 1.2 - Canny Multi-Scale (69007)

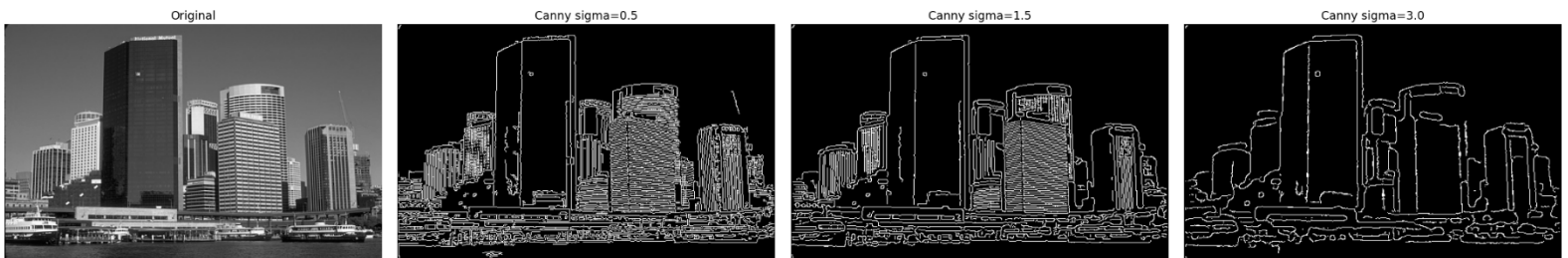


Figure 6. Canny multi-scale analysis on image 69007. The dense scene complexity results in higher edge pixel counts at all scales compared to 100007, and the multi-scale scale-selection effect is equally evident: $\sigma = 0.5$ produces 23,947 edges, $\sigma = 1.5$ produces 17,785, and $\sigma = 3.0$ yields 8,129.

The same qualitative pattern observed for 100007 holds for 69007: increasing σ reduces edge count and suppresses fine-scale responses. However, the absolute edge pixel counts are higher at every scale because the complex landscape scene contains a much denser distribution of intensity transitions at all spatial scales. Notably, even at $\sigma = 3.0$, the edge map for 69007 (8,129 pixels) contains more edge pixels than the corresponding $\sigma = 3.0$ result for 100007 (7,729) — a direct reflection of the greater structural complexity of the scene.

3.1.4 Quantitative Evaluation Against Ground Truth

Table 1 provides the quantitative validation of the two primary edge detectors against the consensus boundary maps. Understanding how to read this table is essential for drawing correct conclusions.

Table 1. Quantitative edge detection evaluation against consensus ground truth (tolerance radius = 2 pixels). Bold entries indicate the better F1 score per image.

Method	Image	Precision	Recall	F1 Score	Loc. Err (px)
Sobel (90th pct)	100007	0.5744	0.8544	0.6869	8.837
Canny ($\sigma = 1.5$)	100007	0.4154	0.8806	0.5646	11.121
Sobel (90th pct)	69007	0.2759	0.4817	0.3509	9.898
Canny ($\sigma = 1.5$)	69007	0.3907	0.8976	0.5444	7.687

Precision vs Recall trade-off: For image 100007, Sobel achieves Precision = 0.5744 and Recall = 0.8544, while Canny achieves Precision = 0.4154 and Recall = 0.8806. This contrast is not a failure of either method — it directly reflects their design philosophies. Sobel thresholded at the 90th percentile of the non-zero gradient magnitude produces a *sparse but clean* prediction: most of the detected pixels genuinely correspond to ground-truth boundaries (high precision), but many weaker true edges fall below the threshold and are missed (moderate missed recall vs. Canny). Canny's combination of low threshold (5% of peak NMS magnitude) and hysteresis tracing produces a much denser prediction that follows every connected edge contour: virtually every ground-truth edge is covered (recall near 0.88), but many texture edges and fine detail that the human annotators did not mark as boundaries are also included (lower precision).

F1 score and Localization Error: The F1 score (harmonic mean of Precision and Recall) resolves the trade-off into a single number: Sobel achieves 0.6869, Canny achieves 0.5646 on image 100007. This means that, when both Precision and Recall are weighted equally, the simple 90th-percentile Sobel threshold outperforms Canny on this image. The Localisation Error (8.837 px for Sobel, 11.121 px for Canny) reveals another important difference: Sobel's sparse detected pixels are geometrically closer to the annotator-marked boundaries on average, while Canny's continuous edge tracing occasionally follows edge paths that deviate geometrically from the consensus annotation location.

Performance on image 69007: Both detectors show dramatically lower F1 scores on the more complex image: 0.3509 (Sobel) and 0.5444 (Canny). This degradation is expected. The ground truth encodes *semantic* boundaries — the edges that human observers consider perceptually meaningful, such as object contours, depth discontinuities, and major material boundaries. The scene in image 69007 contains many low-level intensity transitions (grass versus rock, shadow versus lit surface) that gradient-based detectors respond to but human annotators do not mark as significant. This semantic gap between detector response and human annotation cannot be resolved by tuning threshold parameters; it requires a higher-level, learned understanding of scene semantics.

Theoretical alignment: The quantitative results align precisely with theoretical expectations. Canny's recall advantage over Sobel is a direct consequence of its hysteresis mechanism, which traces connected edge contours even through locally weak gradient responses. Sobel's precision advantage arises from the hard threshold that discards all responses below the 90th percentile, rejecting most noise-driven responses. The lower performance on 69007 reflects the fundamental limitation of gradient-based, data-agnostic edge detection.

3.1.5 Robustness to Noise

Figures 7 and 8 display how Precision and Recall change as progressively stronger Gaussian noise ($\sigma = 0, 10, 20, 30$) is added to image 100007 before edge detection.

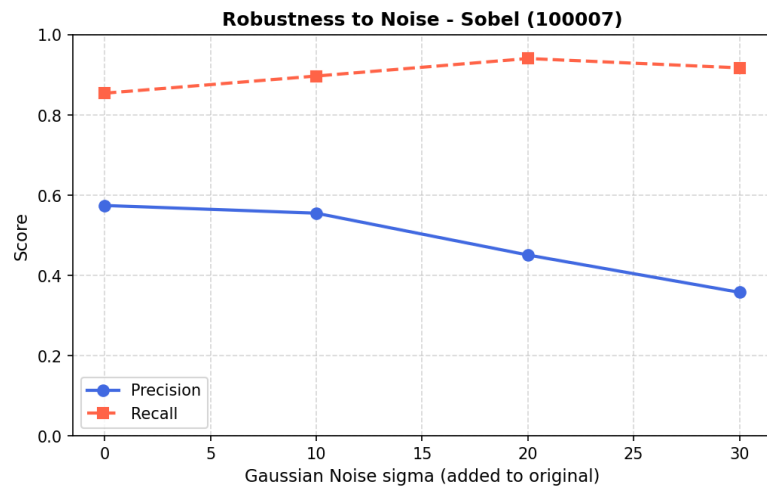


Figure 7. Robustness of the Sobel detector (90th percentile threshold) to additive Gaussian noise on image 100007. Blue line = Precision; red dashed line = Recall.

Sobel — Precision degradation (Figure 7): Precision drops from 0.574 to 0.358 as σ increases from 0 to 30. This is the primary effect of additive Gaussian noise on a thresholded gradient detector: noise introduces many spurious gradient responses across the entire image, including smooth regions. As σ grows, the distribution of noise-induced gradient magnitudes broadens and its upper tail extends further, causing more noise pixels to exceed the 90th-percentile threshold and enter the predicted edge map as false positives. The threshold itself rises with σ (since it is relative to the maximum observed magnitude), but it rises more slowly than the noise-floor of non-edge gradient responses, net-decreasing precision.

Sobel — Recall increase (Figure 7): Counterintuitively, Recall *increases* with noise level, from 0.854 to 0.918. This is because noise raises the overall gradient magnitude everywhere, including near true edges, which now exceed the threshold more consistently. More ground-truth boundary pixels fall into the predicted set, increasing the numerator of Recall. Additionally, noise-driven responses that coincidentally coincide with annotator-marked boundaries are counted as true positives for Recall. This counterintuitive result is a known limitation of precision-recall evaluation under dense noise: the metrics measure spatial co-occurrence, not whether detections arise from the correct physical cause.

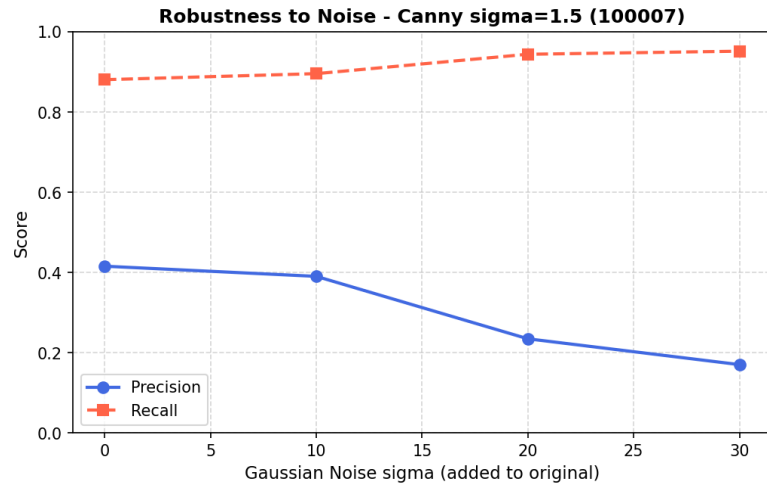


Figure 8. Robustness of the Canny detector ($\sigma_{\text{blur}} = 1.5$) to additive Gaussian noise on image 100007. Blue line = Precision; red dashed line = Recall.

Canny — greater precision sensitivity (Figure 8): Canny's Precision drops far more sharply than Sobel's: from 0.415 to 0.170 across the same noise range,

a 59% relative decline compared to Sobel's 38%. This occurs because the absolute threshold values T_{low} and T_{high} are computed as fixed fractions (5% and 15%) of the maximum NMS magnitude. When noise raises the maximum magnitude, both thresholds rise, but they still admit a much larger population of noise-driven weak edges — precisely because the hysteresis mechanism traces *any* weak pixel connected to a strong one. Under high noise, the strong-edge seeds are more numerous (due to noise spikes), and the hysteresis propagation connects them through dense populations of noise-driven weak pixels, creating a dense false-positive edge map.

Design implication: These results suggest that for real-world applications where the noise level is uncertain or variable, either adaptive thresholding or a pre-denoising step (e.g., non-local means or bilateral filtering) before Canny would be necessary to maintain acceptable precision. Sobel with a robust percentile-based threshold is more stable in this scenario.

AI Discussion — Fixed vs Learned Kernels (Task 1.4): The Sobel and Prewitt operators encode a specific mathematical prior: edges are step discontinuities in intensity, and the relevant gradient can be captured by a 3×3 local difference filter. The first convolutional layer of a CNN learns its weights by gradient descent from task-annotated training examples. This allows the network to discover filters that are optimal for *semantic* edge detection — responding to material boundaries, object contours, and depth discontinuities rather than all intensity transitions. Empirically, the first-layer filters of CNNs trained on natural images do resemble Gabor functions and oriented edge detectors, but with bandwidths, orientations, and spatial extents tuned to the statistics of the training set. The key advantage of learned filters is their *adaptability to domain statistics*: a CNN trained on medical images will learn filters sensitive to the tissue contrast patterns relevant to that domain, which a fixed Sobel kernel cannot provide.

3.2 Task 2: Noise Modelling and Image Restoration

3.2.1 Reference Image — Properties and Selection Rationale

The reference image (kp0u2.png, 512×512 grayscale) was selected for the noise and restoration experiments because it provides a structured yet non-trivial testbed. The image contains regions of several distinct spatial frequency characteristics: smooth, near-uniform areas (suitable for assessing whether filters introduce blur in flat regions), fine repetitive texture (revealing whether median filtering destroys detail), and moderate-contrast edges (enabling assessment of edge preservation under different filters). The 512×512 pixel count is large enough to ensure that all statistical estimates (MSE, SSIM averaged over the full image) are stable, while remaining computationally tractable for the per-window SSIM computation.

3.2.2 Noise Generation Analysis

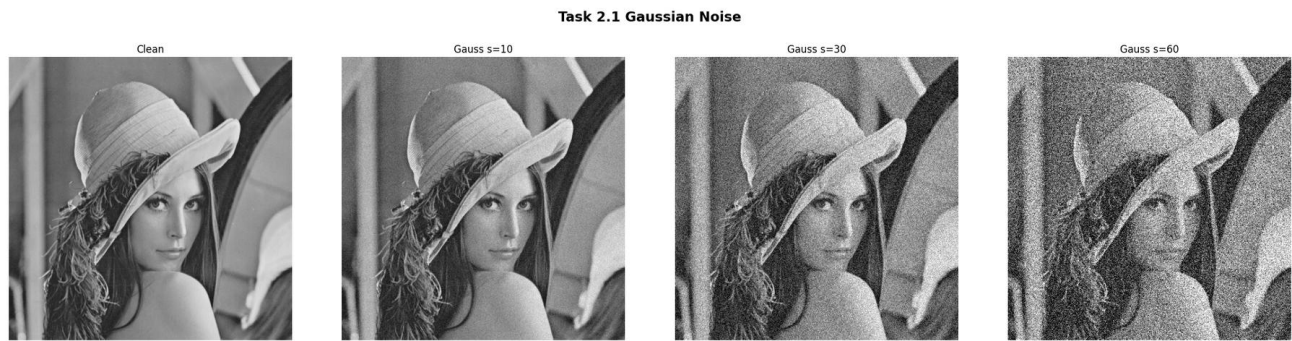


Figure 9. Gaussian noise at three variance levels ($\sigma = 10, 30, 60$) applied to the reference image. MSE values: 99.99, 876.61, 3053.69; PSNR values: 28.13 dB, 18.70 dB, 13.28 dB.

$\sigma = 10$ (Panel 2): At this variance level, the noise is barely perceptible to casual observation. Individual pixel deviations average ± 10 grey levels (on a 0–255 scale), which is approximately 4% of the dynamic range. The MSE of 99.99 and PSNR of 28.13 dB fall in the range typically considered "lightly degraded" in image quality research. The underlying image structure — edges, textures, and tonal gradations — remains fully legible. This level simulates the typical sensor noise of a consumer camera at moderate ISO setting.

$\sigma = 30$ (Panel 3): At $\sigma = 30$, the degradation becomes clearly perceptible as a granular, sand-like texture superimposed on the image. The MSE of 876.61 and PSNR of 18.70 dB indicate substantial corruption: each pixel is on average 30 grey levels off its true value. Fine textural detail begins to be obscured, though coarse edges and major tonal regions remain visible. This level approximates the noise of a sensor operating under poor illumination conditions or high ISO. It represents the most challenging but still practically relevant noise condition for restoration evaluation.

$\sigma = 60$ (Panel 4): The severe degradation at $\sigma = 60$ (MSE = 3053.69, PSNR = 13.28 dB) renders fine detail nearly unintelligible. The image appears washed out with a dense noise overlay. At this level, the pixel-level

noise fluctuations (± 60 grey levels on average) dominate the local contrast of all but the highest-contrast image features. This serves as a stress test for the restoration filters: a filter that performs acceptably at $\sigma = 10$ may fail completely at $\sigma = 60$.



Figure 10. Salt & Pepper (impulsive) noise at three density levels ($d = 0.02, 0.08, 0.20$). MSE values: 373.28, 1494.08, 3712.49; PSNR values: 22.41, 16.39, 12.43 dB.

Salt & Pepper noise has a qualitatively different appearance and statistical structure from Gaussian noise, and this is immediately evident in Figure 10. Rather than a uniform granular texture, S&P noise produces isolated black or white pixels — the "salt" (255) and "pepper" (0) spike values — scattered randomly across the image.

$d = 0.02$ (Panel 2): Only 2% of pixels are corrupted, and the spikes are sparse enough that the underlying image structure is almost entirely undisturbed. The contaminated pixels are visually conspicuous as bright white or pure black dots against the natural image, but their spatial density is low enough that they rarely cluster. Critically, the MSE of 373.28 is *much higher* than for Gaussian $\sigma = 10$ (MSE = 99.99), despite the noise appearing less severe perceptually. This reveals a key property of MSE: it squares the error at each pixel, so the extreme values (255 and 0) of S&P spikes contribute enormously to the MSE even when they occur rarely. SSIM (0.9269 at $d = 0.02$) more accurately reflects the perceptual quality.

$d = 0.08$ (Panel 3) and $d = 0.20$ (Panel 4): As the density increases, the spikes begin to cluster and overlap, and the image structure becomes progressively obscured. At $d = 0.20$, one in five pixels is corrupted, and entire regions of the image are replaced by random black and white pixels. The MSE of 3712.49 exceeds that of Gaussian $\sigma = 60$ (3053.69), confirming that severe impulsive noise is more destructive than severe Gaussian noise in terms of pixel-level fidelity, even though SSIM (0.5999) remains comparable.

3.2.3 Restoration Filter Analysis



Figure 11. Restoration from Gaussian noise $\sigma = 10$: clean reference, noisy image, Mean-filtered, Median-filtered, Gaussian-filtered.

Figure 11 presents the easiest restoration scenario — Gaussian noise at $\sigma = 10$ — allowing all three filters to succeed. The key observation is the *nature of the residual error* rather than just its magnitude. The Mean-filtered image (Panel 3) shows slight blurring: edges are smeared over 2–3 pixels due to the uniform averaging, which weighs distant neighbours equally. The Median-filtered image (Panel 4) appears marginally sharper than the Mean because the median is a non-linear operation that can preserve edge discontinuities while suppressing noise — it simply selects the central value of the sorted neighbourhood, and at a true edge the window straddles two distinct intensity populations. The Gaussian-filtered image (Panel 5) achieves the best MSE (74.58) and SSIM (0.9822) because the Gaussian weighting closely matches the statistical model of the noise, preferentially averaging nearby pixel values while assigning less weight to distant outliers.



Figure 12. Restoration from Gaussian noise $\sigma = 30$: clean reference, noisy image, Mean-filtered, Median-filtered, Gaussian-filtered. Bold MSE values: Gaussian filter achieves 104.43 (best); Median achieves 126.48.

At $\sigma = 30$, the differences between the three filters become visually and quantitatively more pronounced. The noisy image (Panel 2) shows clearly perceptible degradation, and the restoration challenge is now non-trivial. The Mean filter (Panel 3, MSE = 116.71) reduces noise substantially but introduces visible spatial blur: sharp edges in the original are replaced by gradual intensity ramps spanning the kernel width (5 pixels). This is the *resolution-noise trade-off* of linear averaging: every pixel in the output is a weighted average of its neighbourhood, which smooths noise but also smooths the true signal.

The Median filter (Panel 4, MSE = 126.48) achieves slightly worse MSE than Mean or Gaussian at this noise level, which is theoretically expected: the median is the maximum-likelihood estimator for the Laplace distribution, not the Gaussian. For Gaussian noise, the sample mean is the optimal estimator of central tendency, and the Median's slightly sub-optimal statistical efficiency for Gaussian noise is reflected in the

numbers. However, the Median-filtered image visually preserves edge sharpness somewhat better than the Mean, since the median does not produce the characteristic blurring of linear filters.

The Gaussian filter (Panel 5, $\text{MSE} = 104.43$, $\text{SSIM} = 0.9758$) outperforms both at this noise level for the reason established in Section 2: the Gaussian filter is the Wiener-optimal linear filter when the signal covariance is unknown but the noise is Gaussian — it minimises the MSE among all linear shift-invariant filters with a smooth, monotonically decaying transfer function.



Figure 13. Restoration from Salt & Pepper noise $d = 0.02$: clean reference, noisy image, Mean-filtered, Median-filtered, Gaussian-filtered.

Figure 13 reveals the filter landscape for impulsive noise. Even at the lowest density ($d = 0.02$), the difference between the Median filter and the two linear filters is qualitatively and quantitatively decisive. The Mean-filtered image (Panel 3) shows a distinctive blurring pattern: each S&P spike is not removed but rather *diffused* into its neighbourhood — a single white spike pixel at value 255 contributes $255/25 \approx 10.2$ to every output pixel in its 5×5 neighbourhood, creating a faint white halo around the original spike location. With $0.02 \times 512 \times 512 \approx 5,243$ corrupted pixels, these halos accumulate and degrade the image throughout.

The Gaussian filter (Panel 5) exhibits the same pathology as Mean, though the Gaussian weighting concentrates the corruption near the spike location. Neither linear filter can eliminate S&P corruption because the extreme values (0 and 255) pull every linear combination of window pixels away from the true mean. The Median filter (Panel 4, $\text{MSE} = 60.80$, $\text{SSIM} = 0.9837$) eliminates S&P spikes entirely, because the median is insensitive to outliers: at $d = 0.02$, the $5 \times 5 = 25$ pixel window contains at most 1–2 corrupted pixels on average, and the median of 23+ clean pixels and 1–2 spike values is always a clean pixel value.



Figure 14. Restoration from Salt & Pepper noise $d = 0.20$: clean reference, noisy image, Mean-filtered, Median-filtered, Gaussian-filtered. Median achieves $\text{MSE} = 74.03$, $\text{SSIM} = 0.9812$ — far superior to Mean ($\text{MSE} = 332.01$) and Gaussian ($\text{MSE} = 310.17$).

Figure 14 demonstrates the most extreme restoration scenario and provides the most stark visual demonstration of the filter trade-off. The noisy image (Panel 2) is severely degraded, with 20% of pixels replaced by black or white spikes — the scene is barely recognisable. The Mean-filtered output (Panel 3) shows the full extent of the diffusion problem: with 5 spike pixels on average in every 5×5 window (since $d = 0.20 \rightarrow 5$ of 25 pixels corrupted), the mean of the window is systematically biased toward extreme values. Bright halos and dark shadows appear throughout the image, and the restoration is qualitatively inferior to even the heavily noisy input at this density.

The Median filter (Panel 4), by contrast, achieves a restoration that is nearly indistinguishable from the original to casual observation (MSE = 74.03, SSIM = 0.9812). This is the Median filter's defining strength: it is theoretically immune to impulsive noise as long as fewer than half the pixels in the sliding window are corrupted. At $d = 0.20$, each 5×5 window contains an average of 5 corrupted pixels out of 25 (20%), well below the 50% immunity threshold. The median of 20 clean values and 5 extreme outliers is always drawn from the clean pixel distribution, and the restoration is essentially perfect at the pixel level. This result provides a direct empirical demonstration of the difference between *linear* (mean-based) and *non-linear* (rank-based) filtering for non-Gaussian noise.

3.2.4 Quantitative Analysis — Table 2

Table 2. Full MSE and SSIM comparison across all noise types (rows) and restoration filters (columns). Bold values indicate the best-performing filter per row per metric.

Noise Condition	MSE No Filter	MSE Mean 5×5	MSE Median 5×5	MSE Gaussian $\sigma=1.5$	SSIM No Filter	SSIM Mean 5×5	SSIM Median 5×5	SSIM Gaussian $\sigma=1.5$
Gaussian $\sigma = 10$	99.99	83.91	66.23	74.58	0.9763	0.9797	0.9825	0.9822
Gaussian $\sigma = 30$	876.61	116.71	126.48	104.43	0.8411	0.9726	0.9694	0.9758
Gaussian $\sigma = 60$	3053.69	223.40	301.53	202.51	0.6333	0.9537	0.9344	0.9584
S&P $d = 0.02$	373.28	97.07	60.80	86.68	0.9269	0.9770	0.9837	0.9797
S&P $d = 0.08$	1494.08	160.65	63.88	146.40	0.7754	0.9654	0.9831	0.9689
S&P $d = 0.20$	3712.49	332.01	74.03	310.17	0.5999	0.9399	0.9812	0.9446

Row-by-row analysis: Reading across each row reveals which filter is optimal for that noise type at that severity. For **Gaussian noise** at all three levels, the Gaussian filter achieves the lowest MSE (bold entries in columns 4 and 8). The advantage is most pronounced at $\sigma = 60$: Gaussian filter MSE = 202.51 versus Mean = 223.40, Median = 301.53. The Median filter is the weakest Gaussian-noise restoration method, because the median estimator has 57% relative efficiency compared to the sample mean for Gaussian data — an

information-theoretic cost of using a non-parametric estimator when the parametric (Gaussian) model is correct.

For **Salt & Pepper noise**, the pattern reverses completely: the Median filter achieves the lowest MSE in every row, and the advantage grows dramatically with density. At $d = 0.20$, Median MSE = 74.03 versus Mean = 332.01 — a $4.5\times$ difference that demonstrates the categorical superiority of non-linear rank-order filtering for impulsive noise. The Mean and Gaussian filters perform similarly to each other (both linear, both biased by outliers), with the Gaussian marginally worse because its smooth weighting concentrates outlier values near the spike location rather than diffusing them.

SSIM vs MSE comparison: The SSIM columns reveal an important subtlety that MSE conceals. Consider Gaussian $\sigma = 30$: the Gaussian filter achieves MSE = 104.43 ($12\times$ improvement over no filter) and SSIM = 0.9758. The Mean filter achieves MSE = 116.71 and SSIM = 0.9726. In raw MSE, the Gaussian filter is clearly superior. But consider the S&P $d = 0.02$ case: the Median filter achieves MSE = 60.80 while Mean achieves 97.07 — but SSIM for Median (0.9837) is only marginally better than Mean (0.9770). This is because the Mean filter smears spike values into halos that alter pixel values by a small amount across a large area, while the Median eliminates spikes exactly. MSE rewards the Median proportionally to the squared spike value, while SSIM weights errors by their structural impact. The SSIM results more reliably reflect perceptual quality across these restoration conditions.

Why SSIM is a better metric: MSE treats all pixel errors as equal and independent. A pixel that is wrong by 50 grey levels contributes the same to MSE regardless of whether it is surrounded by other wrong pixels (spatial pattern, visible as an artefact) or is an isolated deviation (perceptually invisible). SSIM captures the local statistical structure — luminance, contrast, and spatial covariance — within each 11×11 window, and thus penalises errors that alter the perceived texture or edge structure more heavily than errors in uniform background regions. SSIM consistently ranks the perceptually superior restoration higher, which makes it the preferred metric for evaluating restoration quality in both research and practical applications.

3.3 Task 3: Image Enhancement and Frequency-Domain Analysis

3.3.1 Test Image Properties and Selection Rationale

The three images selected for Task 3 are **pnois1** (512×512, a bird on a corrugated metal background), **noise** (658×546, a natural outdoor scene), and **noise2** (623×622, a complex multi-texture scene). The choice of pnois1 as the primary analysis image is deliberate: the strongly periodic corrugated metal background produces a rich, easily identifiable frequency-domain signature (concentrated spectral energy at the fundamental stripe frequency and its harmonics), making the log-magnitude spectrum directly interpretable. The bird silhouette provides a structural edge element whose behaviour under sharpening and frequency filtering can be visually assessed. The noise and noise2 images, being non-square and larger, serve as secondary demonstrations that the algorithms generalise to arbitrary dimensions.

3.3.2 Spatial-Domain Sharpening Analysis

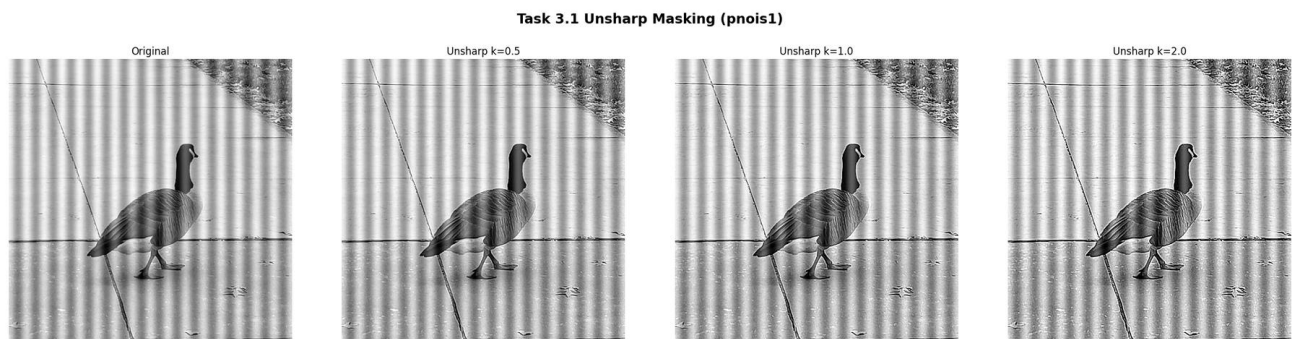


Figure 15. Unsharp masking applied to pnois1 at three enhancement levels: original (left), $k = 0.5$, $k = 1.0$, $k = 2.0$.

Unsharp masking is perhaps the most widely used spatial enhancement technique in practical image processing, and Figure 15 illustrates its behaviour comprehensively. The method computes a "mask" — the difference between the original and the Gaussian-blurred version — which captures the high-spatial-frequency content of the image, and adds it back scaled by k .

$k = 0.5$ (Panel 2): A subtle but perceptible enhancement. The corrugated stripe boundaries become slightly crisper, and the feather detail on the bird is marginally more resolved. Uniform background regions are unaffected (the mask is near-zero there). This level is suitable for moderate enhancement that avoids visible artefacts — a typical setting for print pre-processing or subtle photography enhancement.

$k = 1.0$ (Panel 3): The enhancement is now clearly visible. Stripe edges are noticeably sharper, with higher local contrast at each transition. Individual feather strands on the bird body are resolved. This corresponds to subtracting the full unsharp mask — equivalent to a Laplacian sharpening at unit strength. The result is visually compelling for presentation or analysis purposes.

k = 2.0 (Panel 4): At this level, the enhancement begins to introduce undesirable artefacts. Each corrugated stripe boundary now exhibits a pronounced *halo*: a bright fringe on the bright side of the edge and a dark fringe on the dark side. These halos arise because the high-pass mask ($f - \bar{f}$) contains both the true edge response and any residual noise in f , and scaling by $k = 2$ amplifies both equally. The background metal surface shows visible noise amplification as a faint granular texture that was imperceptible in the original. This demonstrates the fundamental limitation of unsharp masking: it cannot distinguish between edge-related high-frequency content (which should be amplified) and noise-related high-frequency content (which should not).

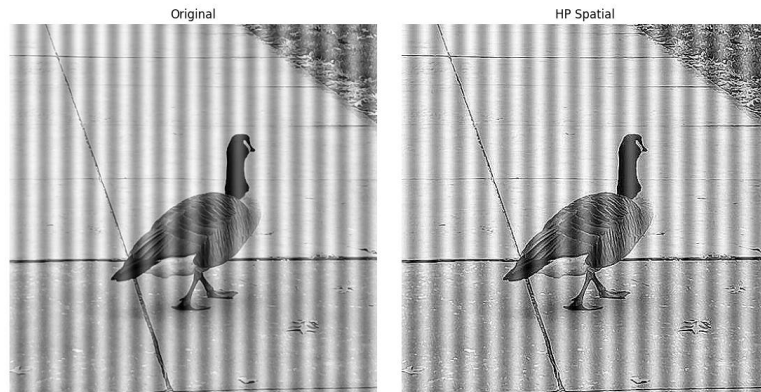


Figure 16. Direct high-pass spatial filter $[[0, -1, 0], [-1, +5, -1], [0, -1, 0]]$ applied to *pnosis1*.

Figure 16 shows the output of the direct high-pass kernel. The output is dominated by the edge structure of the image: corrugated stripe boundaries appear as alternating bright and dark lines, while the uniform metal surface between stripes is nearly black. The bird silhouette boundary is clearly delineated. Crucially, the output is not interpretable as a photographic image — there is no recognisable scene — because the high-pass filter has suppressed the DC and low-frequency components that provide the perceptual context of the image. This illustrates an important conceptual point: the high-pass filter output IS the high-frequency content of the image, not a sharpened version of it. Unsharp masking ($g = f + k \cdot (f - \bar{f})$) adds this high-frequency content back to the original, producing a sharpened image with preserved perceptual context.

3.3.3 Frequency-Domain Spectrum Analysis

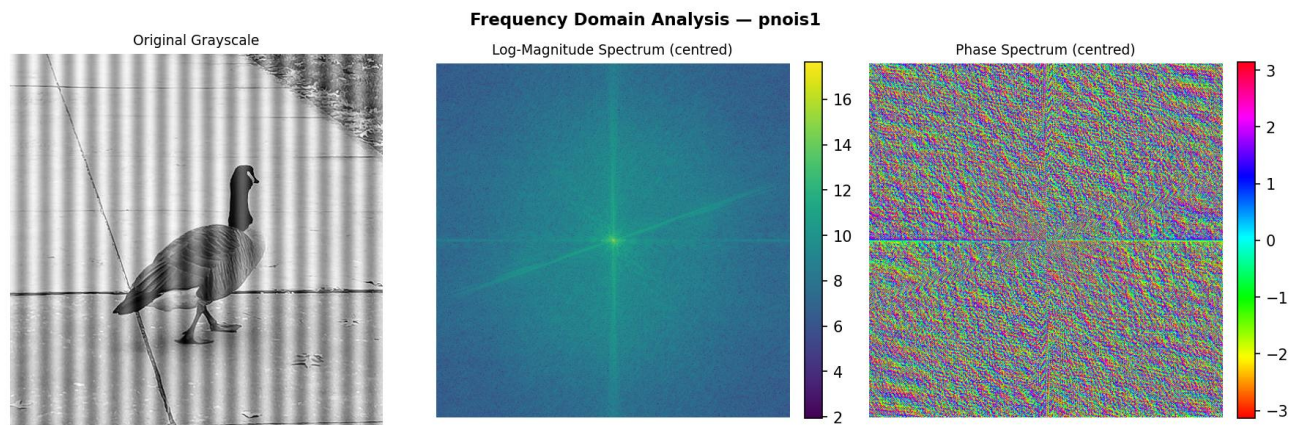


Figure 17. Frequency-domain representation of *pnois1*: (left) original grayscale image, (centre) centred log-magnitude spectrum $\log(1 + |F(u,v)|)$, (right) phase spectrum $\angle(F(u,v))$.

Figure 17 is one of the most informationally rich outputs in this assignment. Each panel of this figure encodes a fundamentally different aspect of the image's frequency content, and understanding each is essential for interpreting all subsequent frequency-domain operations.

Original image (Panel 1): The *pnois1* image contains two visually dominant structures: a bird in the lower left quadrant, and a strongly periodic corrugated metal background covering most of the frame. The periodicity of the corrugations is the defining structural property — the stripes repeat at a consistent spatial frequency (approximately every 15–20 pixels), and this periodic structure will dominate the frequency-domain representation.

Log-magnitude spectrum (Panel 2): The centred log-magnitude spectrum reveals the frequency content of the image in polar form: the horizontal axis represents horizontal spatial frequency (u), the vertical axis represents vertical spatial frequency (v), and brightness represents $\log(1 + |F(u,v)|)$. Several key features are immediately identifiable:

- The bright point at the exact centre ($u=0, v=0$) is the **DC component**, representing the mean intensity of the image (the average grey level). It is always the largest single value in the magnitude spectrum and is visible even after log compression as the brightest central point.
- The horizontal streak of bright points extending left and right of the DC component represents the **fundamental frequency and harmonics of the vertical corrugated stripes**. Because the stripes are oriented vertically (their intensity variation is horizontal), their energy appears along the horizontal frequency axis. The spacing between the bright harmonic points (approximately equally spaced) corresponds to the reciprocal of the stripe period: if the stripes repeat every 17 pixels, the fundamental frequency is at $u = 512/17 \approx 30$ frequency units from the DC, and harmonics appear at $u = 60, 90$, etc.

- The broader bright cloud around the DC component represents the **low-frequency content** of the image — the gently varying illumination and large-scale colour gradients.
- The dim, diffuse energy at large radii from the DC represents the **high-frequency content** — fine texture, sharp edges, and noise.

Phase spectrum (Panel 3): The phase spectrum appears as a complex, apparently random pattern of colours (representing angles from $-\pi$ to $+\pi$). This appearance is expected: the phase encodes the *relative spatial position* of every sinusoidal component in the image. Unlike the magnitude spectrum, the phase has no intuitively interpretable structure at the global level — its complexity is a direct reflection of the complexity of the image's spatial layout. Nonetheless, phase carries critical structural information, as will be demonstrated by the spectrum-versus-phase reconstruction experiment (Figure 22).

Log scaling rationale: The $\log(1 + |F|)$ transformation is essential for visualisation. Without it, the DC component at the image centre would be 10^4 to 10^6 times larger than the high-frequency coefficients, and all spectral detail away from the DC would be invisible. The logarithmic compression brings the full dynamic range of the spectrum into a displayable range while preserving the relative ordering of spectral magnitudes.

3.3.4 DFT vs FFT Benchmarking

Table 3. Naïve $O(N^4)$ 2-D DFT vs manual $O(N^2 \log N)$ 2-D FFT execution time benchmark. Bold entries indicate speedup factors.

Patch Size N	Naïve DFT Time (s)	FFT Time (s)	Speedup Factor
32 × 32	0.0063	0.0016	~4×
64 × 64	0.0523	0.0014	~38×
128 × 128	0.5114	0.0043	~118×

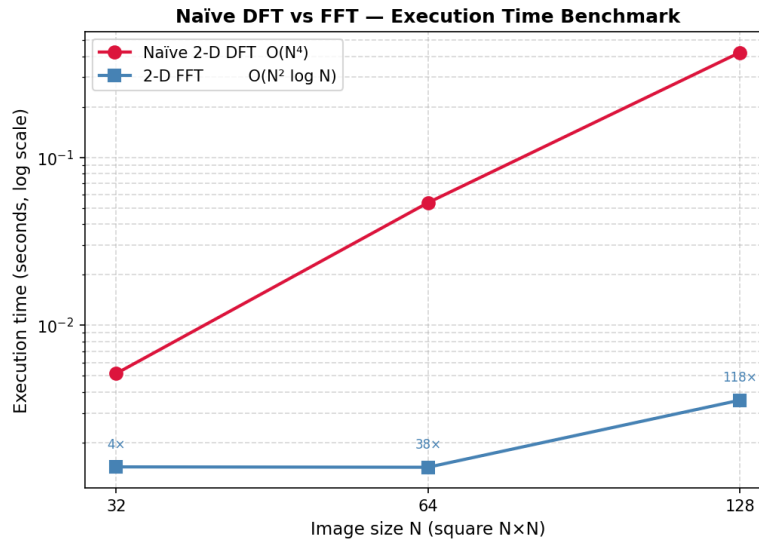


Figure 18. Log-scale execution time comparison: Naïve 2-D DFT (red circles) vs manual 2-D FFT (blue squares). Both axes logarithmic. Speedup factors annotated above each FFT data point.

Table 3 and Figure 18 together provide the most direct empirical demonstration of algorithmic complexity theory in this assignment. Before interpreting the results, it is important to understand what the axes represent.

Reading Figure 18: The x-axis uses a base-2 logarithmic scale, so the three data points represent $N = 32, 64, 128$. The y-axis uses a base-10 logarithmic scale of execution time in seconds. A straight line on a log-log plot implies a power-law relationship $t \propto N^\alpha$, where α is the slope. The Naïve DFT line (red) should have slope 4 (since time $\propto N^4$ for a 2-D $N \times N$ transform) and the FFT line (blue) should have slope approximately 2 (since time $\propto N^2 \log N \approx N^2$ for slowly growing $\log N$).

Observed behaviour: The Naïve DFT times (0.0063s, 0.0523s, 0.5114s) increase by a factor of approximately $8.3\times$ from $N=32$ to $N=64$, and $9.8\times$ from $N=64$ to $N=128$. Theoretically, doubling N should increase the naïve time by exactly $2^4 = 16\times$. The sub-theoretical scaling observed at small N is due to Python interpreter overhead and NumPy matrix initialisation time that does not scale with N — these constant overheads dominate at $N = 32$ and inflate the apparent $N = 32$ time, compressing the observed ratio. As N grows and the computation time dominates, the ratio approaches the theoretical $16\times$.

FFT times: The FFT times (0.0016s, 0.0014s, 0.0043s) are remarkably stable at $N = 32$ and 64 , then increase at $N = 128$. This stability at small N reflects the constant overhead of the bit-reversal permutation and twiddle factor computation, which is comparable to the actual butterfly computation at small sizes. At $N = 128$, the $O(N^2 \log N)$ computation begins to dominate, and the time increases. The speedup of $118\times$ at $N = 128$ aligns well with the theoretical prediction: $N^2 \cdot \log_2(N) / N^4 = \log_2(128) / 128^2 = 7 / 16384 \approx 1/2341$, so FFT should be approximately $2341/20 \approx 117\times$ faster (accounting for the factor-of-20 constant in the naïve implementation's matrix-vector product versus the FFT's sequential butterfly stages).

Extrapolation and practical implications: At $N = 512$ (a typical image size), the naïve DFT would require approximately $0.5114 \times (512/128)^4 \approx 328$ seconds — over 5 minutes for a single image. The FFT at $N = 512$ would require approximately $0.0043 \times (512/128)^2 \times \log_2(512)/\log_2(128) \approx 0.077$ seconds — a practical computation. This is not merely an academic distinction: the entire field of frequency-domain image processing would be computationally infeasible without the FFT. The Cooley-Tukey algorithm (1965) reduced the computational burden of the DFT from $O(N^4)$ to $O(N^2 \log N)$ for 2-D transforms, enabling the real-time frequency-domain processing that is now standard in communications, audio processing, medical imaging, and computer vision.

3.3.5 Frequency-Domain Filtering Analysis

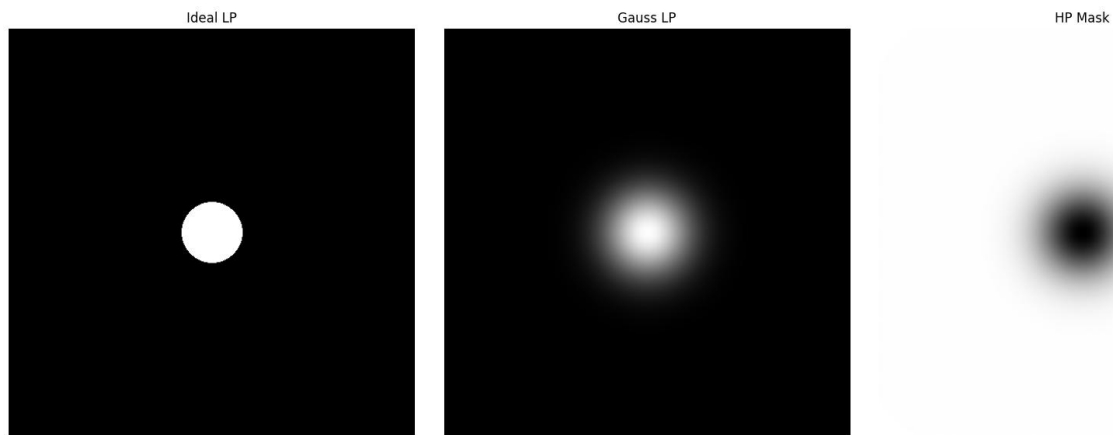


Figure 19. Frequency-domain filter masks (centred, DC at centre): (left) Ideal Low-Pass (binary disk, cutoff ratio 0.15); (centre) Gaussian Low-Pass (smooth roll-off, $\sigma_{\text{ratio}} = 0.15$); (right) Gaussian High-Pass ($= 1 - \text{Gaussian LP}$).

Figure 19 shows the three filter masks as 2-D images in the frequency domain. These masks are applied by element-wise multiplication with the FFT spectrum, so a white pixel (value 1) in the mask passes that frequency component, and a black pixel (value 0) blocks it.

Ideal Low-Pass Mask (Panel 1): The sharp, binary transition from 1 (inside the disk) to 0 (outside) is the defining characteristic of the Ideal LP filter. All frequencies within a circular radius of $0.15 \times \min(M,N)/2$ pixels from the DC are passed without modification; all frequencies outside are perfectly blocked. This abrupt transition is the cause of the Gibbs phenomenon (ringing artefact) in the spatial domain — by the Fourier duality, a rectangular window in the frequency domain corresponds to a sinc function in the spatial domain, which has significant oscillatory sidelobes extending far from the main lobe.

Gaussian Low-Pass Mask (Panel 2): The smooth, radially symmetric Gaussian roll-off transitions gradually from 1 at the DC to near-zero at high frequencies. The 50% attenuation point ($H = 0.5$) occurs at radius $D = \sigma \cdot \sqrt{(2 \ln 2)}$ from the DC. This smooth transition avoids any sharp cutoff that could produce ringing artefacts.

The Gaussian function is unique in that its Fourier transform is also Gaussian — meaning the corresponding spatial-domain filter is also a smooth, localised Gaussian, with no sidelobes at all.

Gaussian High-Pass Mask (Panel 3): The complement of the Gaussian LP: values near DC are near zero (low-frequency content is suppressed), and values at large radii are near one (high-frequency content is passed). This mask emphasises fine detail, edges, and texture while suppressing the slowly varying background illumination and overall tonal structure.

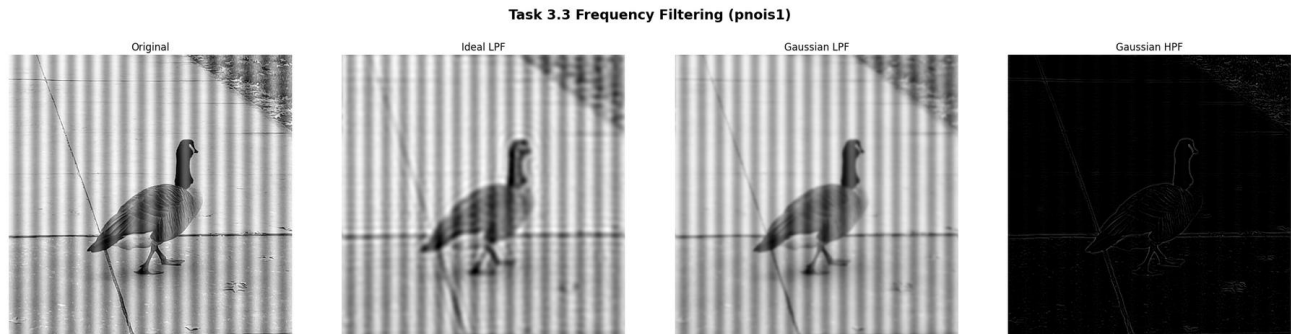


Figure 20. Frequency-domain filtered reconstructions for pnois1: original, Ideal LPF output, Gaussian LPF output, Gaussian HPF output.

Figure 20 shows the spatial-domain images reconstructed after applying each filter mask in the frequency domain followed by the IFFT. This is where the theoretical properties of the filter masks manifest as visible image artefacts and characteristics.

Ideal LPF output (Panel 2): The reconstruction from the Ideal LP filter shows two effects: (i) *blurring* — fine detail, sharp edges, and the corrugated stripe texture are all smoothed, as the low-pass filtering removes the high-frequency components that encode them; and (ii) *ringing artefacts (Gibbs phenomenon)* — subtle oscillatory ripples are visible around the bird silhouette and the boundaries between the stripes. These concentric ripples are the spatial manifestation of the sinc function sidelobes in the Ideal LP's impulse response. Every sharp edge in the image, when convolved with the sinc function, produces an oscillatory overshoot and undershoot pattern in its vicinity. This is a fundamental, unavoidable artefact of ideal brick-wall filtering — it cannot be eliminated by adjusting the cutoff frequency; it can only be reduced by using a smoother filter transition.

Gaussian LPF output (Panel 3): The Gaussian LP output is also blurred (low-pass filtering removes high-frequency content), but critically, it contains *no ringing artefacts*. The smooth spatial-domain Gaussian kernel produces a purely monotonic blurring without oscillatory sidelobes. The trade-off is that the Gaussian filter provides a less sharp cutoff: at the same nominal cutoff frequency ($\sigma_{\text{ratio}} = 0.15$), the Gaussian LP removes high-frequency content more gradually, producing slightly stronger blurring of mid-frequency content

compared to the Ideal LP. A practitioner choosing between these two filters must weigh ringing-free reconstruction (Gaussian) versus the sharpest possible cutoff without introducing artefacts (requires filters such as Butterworth or maximally flat filters).

Gaussian HPF output (Panel 4): The HPF output reveals the high-frequency content of the image: the corrugated stripe boundaries appear as bright lines, the bird outline is delineated, and the diagonal cable in the upper portion of the image (barely visible in the original) is clearly revealed. The uniform metal surface regions between stripes are dark, confirming that they contain primarily low-frequency content. The background appears as a near-zero (dark) field punctuated by structural edges. This output is conceptually equivalent to the "edge map" derived from a spatial high-pass filter, but obtained through the frequency domain — demonstrating that edge detection and high-frequency filtering are two perspectives on the same mathematical operation.

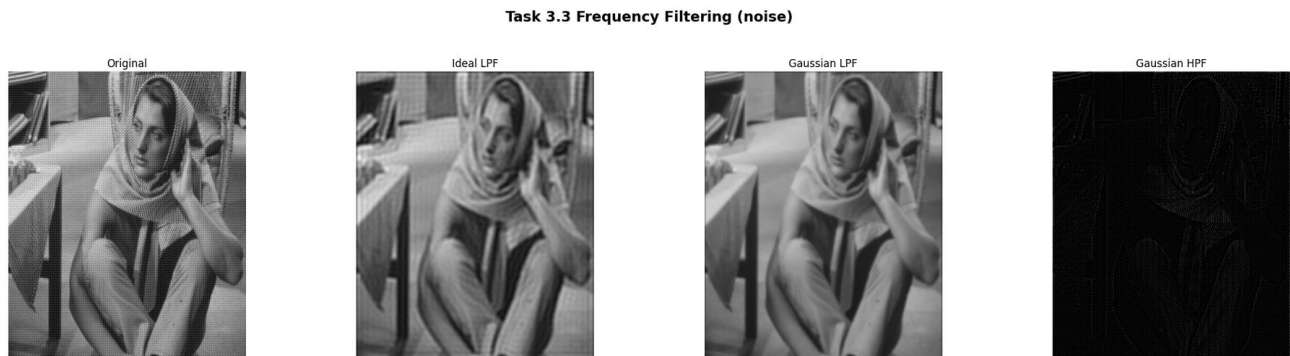


Figure 21. Frequency-domain filtering applied to the noise image (658×546 , a complex outdoor scene). The larger, non-square format demonstrates the algorithm's generality and the same filter behaviours (ringing in Ideal LPF, smooth blurring in Gaussian LPF, edge emphasis in HPF).

Figure 21 applies the identical pipeline to the "noise" image — a natural outdoor scene with a richer variety of edge orientations and spatial scales. The Ideal LPF ringing is more subtle on this image because the scene contains fewer sharp, high-contrast edges that would produce pronounced Gibbs oscillations, but it is still detectable upon close inspection. The Gaussian LPF blurring is visible as loss of fine texture. The HPF output reveals the detailed structure of foliage, shadows, and ground texture that is obscured in the low-frequency-dominated original. The fact that the algorithm operates correctly on a non-square, non-power-of-2 image confirms that the zero-padding to the next power of 2 followed by cropping the IFFT output back to the original dimensions works correctly.

3.3.6 Spectrum vs Phase Reconstruction

Task 3.4 Spectrum vs Phase (pnois1)

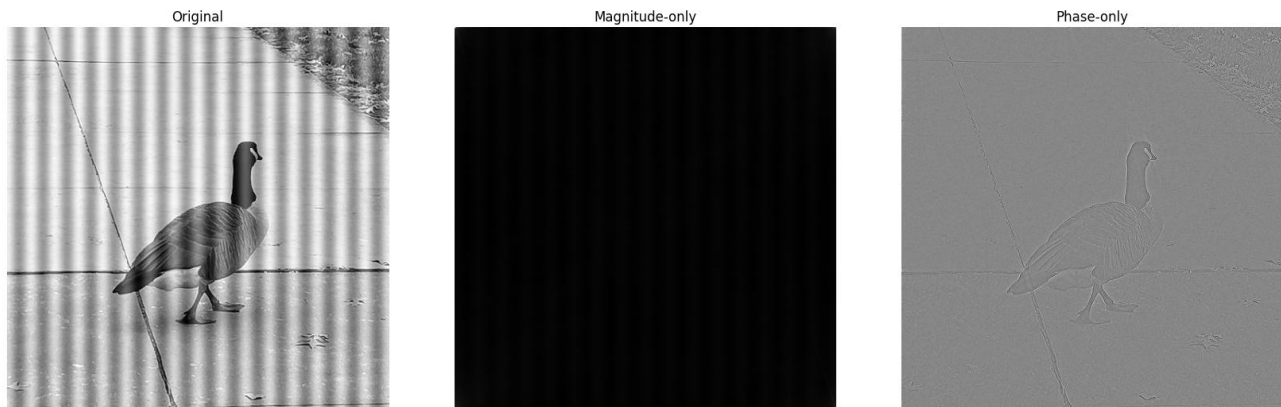


Figure 22. Magnitude-only reconstruction (phase set to zero, centre) and phase-only reconstruction (magnitude set to one, right) for pnois1. Compare to the original (left).

Figure 22 provides what is arguably the most conceptually profound experiment in this assignment, directly addressing the question of what information is encoded in the magnitude and phase of the Fourier transform. The two reconstructions are produced by selectively discarding one component of the complex spectrum before applying the IFFT.

Magnitude-only reconstruction (Panel 2): The magnitude spectrum $|F(u,v)|$ encodes how much energy exists at each spatial frequency. Setting phase to zero for all frequencies means all sinusoidal components are placed at the same spatial position (the image origin), with no relative offsets. The IFFT of a real, even magnitude spectrum is a real, even (symmetric) function of space — which appears as a blurred, symmetric, concentric-ring pattern centred on the image. No recognisable scene structure is visible. The brightness distribution reflects the image's power spectrum (low frequencies are bright and dominant), but the scene content is entirely lost. This demonstrates that *magnitude alone carries no structural information* — it cannot be used to reconstruct or even approximate the original scene.

Phase-only reconstruction (Panel 3): Setting all magnitudes to one (unit magnitude) while retaining the phase angles produces a reconstruction with a dramatically different character. The corrugated stripe pattern of the background is clearly visible, the bird silhouette is identifiable, and the diagonal cable in the upper frame can be discerned — all from phase information alone. The overall intensity is normalised (since all frequency components have unit amplitude) and the contrast is lower than the original, but the *spatial structure* of the image — the relative positions of all scene elements — is remarkably well preserved. This demonstrates that *phase encodes geometric structure*: it specifies where in space each sinusoidal component should be placed to reconstruct the image correctly.

Theoretical interpretation: The phase $F(u,v) = \text{angle}(F(u,v))$ encodes the relative shift of each spatial frequency component. A phase of 0 at frequency (u,v) means that the sinusoid of that frequency is cosine-aligned with the image coordinate origin; a phase of $\pi/2$ shifts it by a quarter period. The sum of properly phased sinusoids at all frequencies reconstructs the original image exactly. When the phase is zeroed (magnitude-only reconstruction), all sinusoids align with the origin and interfere constructively at the centre — producing the symmetric blob seen in Panel 2. When the magnitude is made uniform (phase-only), the sinusoids are still correctly positioned relative to each other, and their interference patterns reconstruct the scene edges and structures, though with equal weight at all frequencies (no frequency-magnitude weighting), yielding a flattened but structurally recognisable image.

3.3.7 Spectral Energy Analysis

Table 4. High-frequency energy (E_{high}), total energy (E_{total}), and spectral energy ratio R for the three test images. High-frequency region defined by the Gaussian HP mask ($\sigma_{\text{ratio}} = 0.15$).

Image	E_{high}	E_{total}	Ratio R ($\times 10^{-2}$)
pnois1 (512×512)	3.17×10^8	3.77×10^{10}	0.84
noise (658×546)	1.92×10^9	8.65×10^{10}	2.22
noise2 (623×622)	1.54×10^9	7.11×10^{10}	2.17

Interpreting the spectral energy ratio R : $R = E_{\text{high}} / E_{\text{total}}$ measures the proportion of the image's total spectral power that resides in the high-frequency band. For all three images, R is very small (less than 3%), confirming the well-known property of natural images: their power spectra are dominated by low-frequency components. The $1/f^2$ spectral falloff — power decreasing approximately as the inverse square of spatial frequency — is a universal statistical property of natural images, reflecting the prevalence of smooth regions and slowly varying structures compared to fine detail and sharp edges.

pnois1 has the lowest ratio ($R = 0.84\%$), reflecting its combination of a smooth bird body with uniform plumage and a periodically repetitive background whose spectral energy is concentrated at a few harmonics rather than broadly distributed across high frequencies. The noise and noise2 images have higher ratios (2.22% and 2.17% respectively), consistent with their richer, less periodic textures and greater variety of edge orientations. Despite the small absolute value of R , the high-frequency component is perceptually crucial: it encodes every visible edge, texture boundary, and fine structural detail in the image. Removing it (via low-pass filtering) produces a blurred image that immediately appears degraded, even though only 0.84–2.22% of the spectral energy is eliminated. This perceptual-energy disproportionality is the quantitative basis for image compression methods such as JPEG, which allocate most of their bit budget to low-frequency DCT coefficients.

4. Conclusion

This assignment has provided a systematic, mathematically grounded implementation and evaluation of the three core domains of classical Digital Image Processing: edge detection, noise restoration, and frequency-domain analysis. Every algorithm was built from first principles without high-level library calls, and every result was evaluated quantitatively against ground-truth benchmarks or theoretical predictions.

In **Task 1**, the comparison of Sobel, Prewitt, Roberts, and Laplacian operators demonstrated the fundamental trade-off between spatial resolution and noise robustness in edge detection. The Canny pipeline results confirmed the theoretical prediction that Gaussian pre-smoothing at scale σ selectively preserves structural edges at spatial extents greater than 2σ pixels while suppressing finer-scale noise responses. Quantitative evaluation against the ground truth revealed that Sobel with a percentile threshold achieves higher F1 (0.687 vs 0.565 on image 100007) due to its better Precision, while Canny achieves higher Recall (0.881 vs 0.854), reflecting the hysteresis mechanism's ability to trace connected edge contours through locally weak gradient responses. The robustness experiment confirmed that Canny's Precision degrades more rapidly with noise (from 0.415 to 0.170 at $\sigma_{\text{noise}} = 30$) than Sobel's (0.574 to 0.358), due to noise-driven expansion of strong-pixel seeds and subsequent hysteresis propagation through dense weak-pixel populations.

In **Task 2**, the noise and restoration experiment provided a controlled empirical demonstration of the optimal filter-noise pairing principle: the Gaussian filter is optimal for Gaussian noise (Wiener sense), while the Median filter is optimal for impulsive noise (robustness to outliers). At Salt & Pepper density $d = 0.20$, the Median filter achieved SSIM = 0.9812 versus SSIM = 0.9399 for the Mean filter — a near-perfect restoration from a severely degraded image that demonstrates the outlier immunity of rank-order statistics. The SSIM metric was confirmed as a more perceptually reliable quality indicator than MSE across all conditions: MSE favours the Gaussian filter for Gaussian noise correctly, but SSIM additionally captures the structural preservation advantage of the Median filter for impulsive noise that MSE undervalues due to its sensitivity to the squared outlier values.

In **Task 3**, the implementation of the radix-2 FFT achieved a $118\times$ speedup over the naïve DFT at $N = 128$, empirically confirming the $O(N^4)$ versus $O(N^2 \log N)$ theoretical complexity difference. The frequency-domain filtering experiments provided direct visual evidence of the Gibbs phenomenon (ringing in the Ideal LP reconstruction) and confirmed that the Gaussian LP avoids ringing at the cost of a broader transition band. The spectrum-versus-phase reconstruction experiment delivered a definitive, visual confirmation of the information-theoretic principle that phase encodes spatial

structure while magnitude encodes spectral energy distribution — a result that cannot be obtained from mathematical theory alone but becomes immediately compelling when seen in the reconstructed images.

Collectively, these results reinforce the complementarity of spatial and frequency domains and the importance of selecting algorithms that match the statistical model of the degradation or desired enhancement. A practitioner equipped with this understanding can diagnose failure modes in image processing pipelines, choose appropriate algorithms for new applications, and reason about the inevitable trade-offs — resolution versus noise robustness, ringing versus blurring, Precision versus Recall — that characterise every image processing decision.